MIL-G- 45668(Ord)
2 May 1960

SUPERSEDING
FCS-827
27 March 1956

MILITARY SPECIFICATION

GENERATOR, PULSE: O-295/M

1. SCOPE

1.1 This specification covers one type of mark generator, hereinafter referred to as "generator". The generator develops a mark and gate to be used in the acquisition indicator display (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U,S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1285

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DRAWINGS

Ordnance Corps

E7620521	Oscillator Audio, Frequency: 0-296/MS
E7620586	Generator, Pulse: 0-295/M
F7620585	Generator, Pulse: 0-295/M Schematic Diagram
F7677411	Resolver, Electrical, Type 23R3N4

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one generator, representative of production, shall be made available to the Government to determine if the generator meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The generator shall be manufactured in accordance with Drawing E7620586, "Generator, Pulse: 0-295/M" and drawings pertaining thereto. The requirements of this specification are detailed only to extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence capabilities prior to final acceptance by the Government.

3.4 Ground potential.- The design and construction of the generator shall insure that all external parts are at ground potential at all times.

3.5 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electron tubes
(f) Workmanship

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3.6 Environmental.

3.6.1 Vibration.- The generator shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second, return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration, the generator shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.6.2 Operating temperatures.- The generator shall meet all the performance requirements of this specification when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.6.3 Storage temperatures.- The generator shall show no evidence of physical failure and shall meet all performance requirements of this specification at the standard ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of 160 degrees and minus 80 degrees Fahrenheit.

3.7 Electrical.

3.7.1 Circuitry.- The circuitry of the generator shall be in accordance with Electrical Schematic, Drawing F7620585.

3.7.2 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the generator are subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than 5 seconds.

3.7.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.8 Performance.- When set-up in accordance with pretest information specified in section 4, the generator shall meet the following requirements. (All symbols referred to in the following paragraphs are those shown on Drawing F7620585.)

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3.8.1 Balance, 4-kilocycle (kc)

3.8.1.1 Potentiometer R20 shall be capable of adjusting for a null at TP1, when the resolver scale is set at zero.

3.8.1.2 The reference carrier component combined with the north-south (N-S) (see figure 1A), shall give a resultant signal which varies as shown on figure 1B.

3.8.1.3 The residual voltage at the null shall not exceed 1 volt.

3.8.2 Azimuth gate.

3.8.2.1 The output voltage of V3B shall vary, when the resolver is rotated through a full revolution (see figure 2).

3.8.2.2 The gate shall have a width 30 degrees (min) at the minus 36-volt points.

3.8.2.3 The gate shall have a width 16 plus or minus 3 degrees at the minus 16-volt points.

3.8.2.4 The gate shall have a minimum positive peak-voltage of 1.0 volt all centered about the resolver null.

3.8.2.5 The peak voltage occurring at approximately 180 degrees shall be negative with a minimum amplitude of 50 volts.

3.8.2.6 The 4-kc component at the cathode of V3B shall not exceed 1.0 volt at any point within the 30 degree area referenced in 3.8.2.2

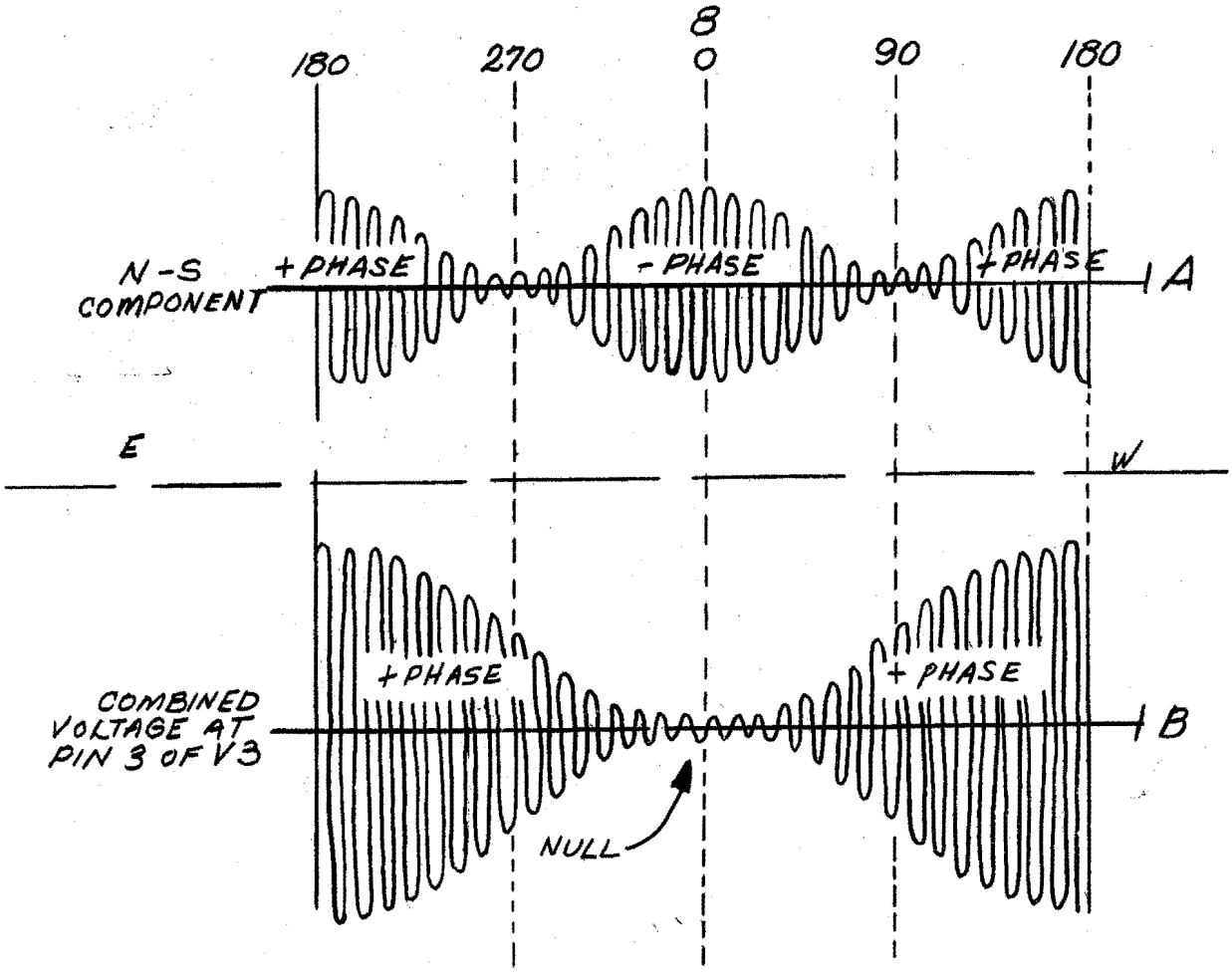
3.8.3 Azimuth mark

3.8.3.1 The output voltage at the suppressor grid of V7 shall vary, when R10 is set fully clockwise and the resolver rotated through a full revolution (see figure 3).

3.8.3.2 The mark gate shall have a width of 3.5 plus or minus 1 degrees at the minus 8-volt points.

3.8.3.3 The peak voltage shall be more positive than minus 1 volt (null point).

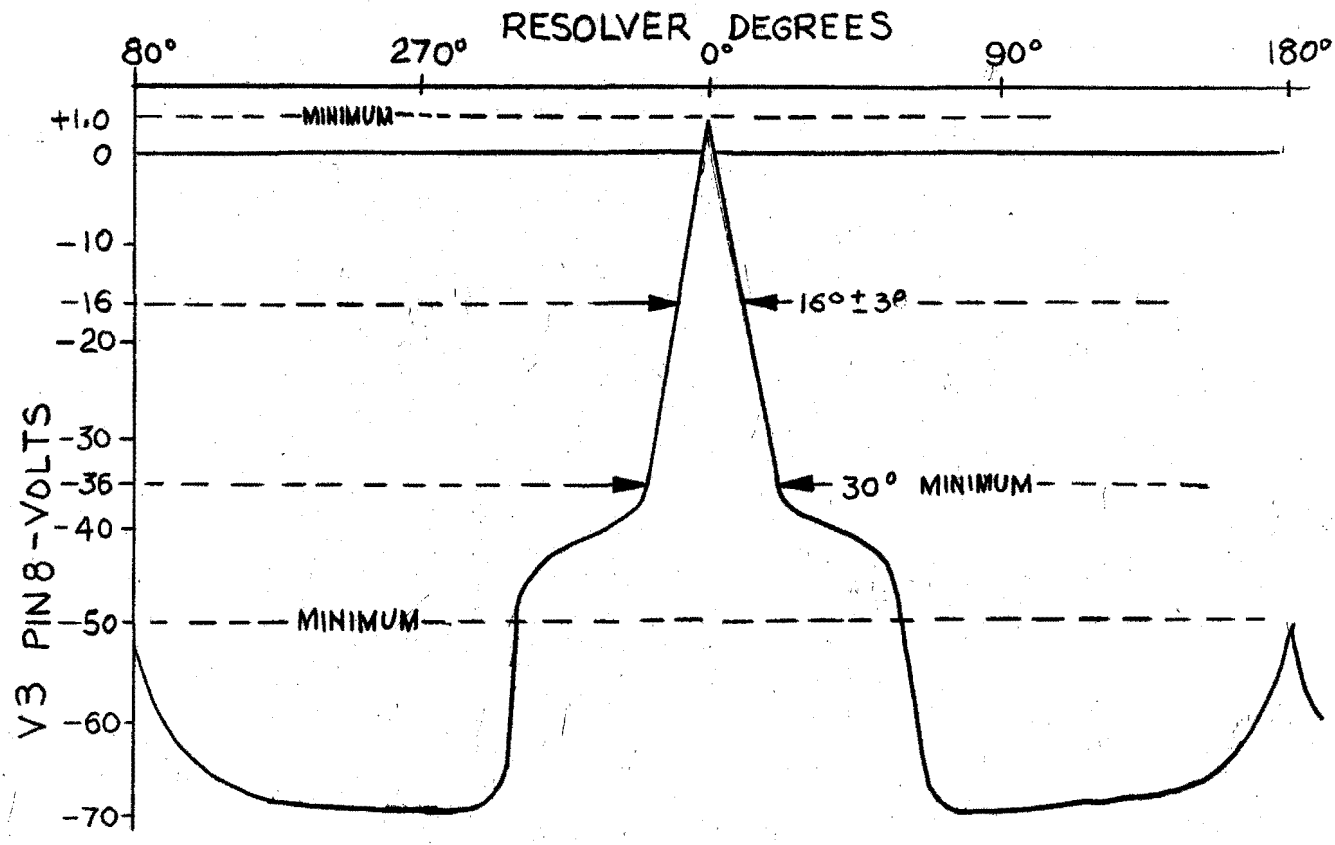
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RESULTANT SIGNAL

FIGURE 1

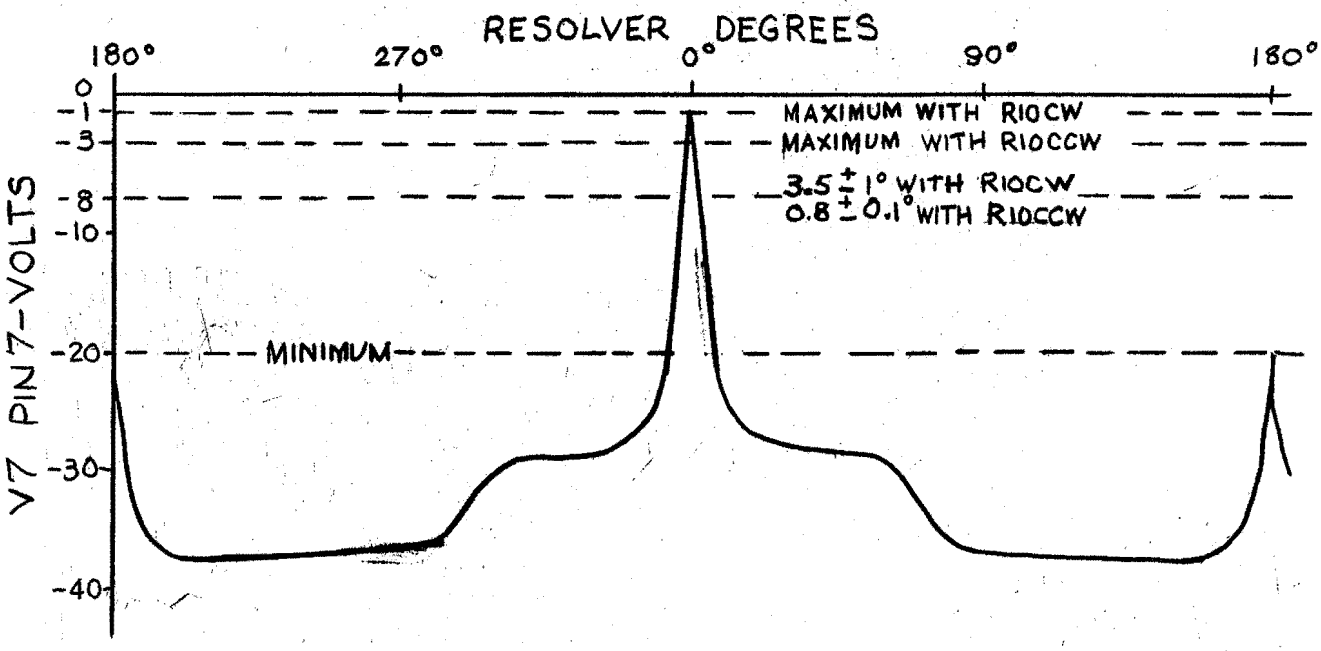
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AZIMUTH GATE

FIGURE 2

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AZIMUTH MARK

FIGURE 3

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3.8.3.4 The peak voltage occurring at approximately 180 degrees shall be negative with a minimum amplitude of 20 volts.

3.8.3.5 The output voltage at the suppressor grid of V7 shall vary, when R10 is set fully counterclockwise and the resolver rotated through a full revolution (see figure 3).

3.8.3.6 The gate shall have a width of 0.8 plus or minus 0.1 degree at the minus 8-volt points.

3.8.3.7 The peak voltage shall be more positive than minus 3 volts at null point.

3.8.4 Azimuth mark triggering level.

3.8.4.1 Multivibrator, V6 shall fire, when the resolver is rotated from minus 3 volts toward 0 volt. The firing shall be indicated by the presence of the azimuth mark pulse at J2.

3.8.4.2 The vacuum tube volt-meter (VTVM) reading between the suppressor grid of V7 and ground, when the multivibrator just fires, shall be negative 10 volts (max).

3.8.5 Azimuth mark repetition rate, length, and amplitude.

3.8.5.1 Potentiometer R32 shall vary the azimuth mark length over a minimum range of 500 to 750 microseconds (u sec).

3.8.5.2 The azimuth mark amplitude shall be a minimum of 6 volts, when the pulse length is 610 plus or minus 5 u sec.

3.8.6 Output at 80 revolution per minute (rpm).- The azimuth mark shall appear at J2 once each revolution of the resolver when rotated 80 plus or minus 5 rpm and R10 set fully clockwise.

3.9 Identification marking. Identification marking unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

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4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

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4.4 Production sample.- Unless otherwise specified in the contract, one generator shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production sample failure.- Should the production sample fail to pass the prescribed examination and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Inspection provisions.

4.5.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.2 Inspection of components.- All positional and functional characteristics shown on the drawints contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the generator. Also included in the above category are all performance requirements specified in notes directly on on the drawings, or in separate performance specifications referenced therein.

4.5.2.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

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TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.5.2.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.6 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.7 Acceptance tests.

4.7.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristcs. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 4.0%	Req't Par.	Q.A. Par.	Test Method
1. Physical failure (post vibration)	3.6.1	4.9.1.1	Visual-Tactile
2. Balance, 4kc	3.6.2	4.9.1.2	Device, Special Testing (DST) Standard Measuring Equipment (SME)
3. Azimuth gate	3.6.2	4.9.1.2	DST- SME
4. Azimuth mark	3.6.2	4.9.1.2	DST- SME
5. Azimuth mark triggering level	3.6.2	4.9.1.2	DST- SME
6. Azimuth mark repetition rate, level, amplitude	3.6.2	4.9.1.2	SME
7. Output at 80 rpm	3.6.2	4.9.1.2	DST-SME

NOTE: The above test shall be conducted at minus 40 degrees and plus 150 degrees Fahrenheit in accordance with the quality assurance paragraph (Q.A. Par.) listed, except the vibration test which shall be conducted at standard ambient temperature (plus 60 degrees to plus 150 degrees Fahrenheit).

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 1.5%	Req't Par.	Q.A. Par.	Test Method
1. Circuitry	3.7.1	4.9.2.1	Standard Meas. Equip. (SME)
2. Dielectric strength	3.7.2	4.9.2.2	SME
3. Insulation resistance	3.7.3	4.9.2.3	SME
4. Balance, 4kc	3.8.1	4.9.3.1	Device, special Testing (DST) SME

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MAJOR: AQL 1.5% (cont'd)	Req't Par.	Q.A. Par.	Test Method
5. Azimuth gate	3.8.2	4.9.3.2	DST-SME
6. Azimuth mark	3.8.3	4.9.3.3	DST-SME
7. Azimuth mark triggering level	3.8.4	4.9.3.4	DST-SME
8. Azimuth mark repetition rate, length, amplitude	3.8.5	4.9.3.5	SME
9. Output at 80 rpm	3.8.6	4.9.3.6	DST-SME
10. Identification marking	3.9	4.9.4	Visual

NOTE: The above test shall be conducted at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit) in accordance with the quality assurance paragraph (Q.A. Par.) listed.

4.8 Inspection equipment.

4.8.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performad by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.8.1.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of 10 percent of the specified tolerance. In cases where the 10 percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

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4.8.1.1.1 Test equipment accuracy requirements.- The following test equipment, or substantially equal, shall be used in accordance with requirements specified in 4.9 to determine the performance requirements specified herein.

Item Number	Description
1.	Voltmeter, d.c., to read plus 5 to minus 80 volts, 20,000 ohms per volt, plus or minus 2 percent.
2.	Electron tube voltmeter, to read 0 to 60 volts d.c., plus or minus 2 percent.
3.	Oscilloscope, TS-239/UP.
4.	Electron tube voltmeter, to read 0 to 1 volt at 4 kilocycles.
5.	Pre-knock pulse, positive 25 volts, 1 to 3 microseconds, 1000 pulses per second.
6.	Generator, 70 volts, 4 kilocycle carrier, constant amplitude and phase, equivalent to Ordnance Corps Drawing E7620521.
7.	Resolver, 28 volts, 4 kilocycles sine and cosine component, equivalent to Ordnance Corps Drawing F7677411.
8.	Power supply, plus 150 volts, 0.010 ampere.
9.	Power supply, minus 250 volts, 0.005 ampere.
10.	Power supply, plus 250 volts, 0.050 ampere.
11.	Power supply, 6.3 volts a-c, 400 cycles.

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4.8.2 Accuracy requirements for devices, special testing.

4.8.2.1 Accuracy requirements for resolver scale (Item 7).- The scale for the resolver shall be so constructed that it may be read within 0.1 degree, through an angle of 5 degrees, centered about the E-W null which occurs when the E-W (terminal I of PI) component changes from the same phase to the opposite phase as the carrier during clockwise rotation of the resolver. The scale shall also be capable of being read within 0.5 degrees over a 60 degree angle centered about this null voltage. The resolver shall be adjusted so that the scale zero and the E-W resolver shall be adjusted so that the scale zero and the E-W resolver output null are coincident.

4.8.2.2 Device to test for vibration requirements.- The device shall consist of a vibrating machine which will produce a motion at a uniformed varied frequency between the limits of 10 and 55 cycles per second. It shall have an amplitude of 0.03 inch (0.06 inch maximum excursion), and shall be equipped with a medium capable of holding the unit securely to the vibrating machine so that motion is applied to the unit in each of the 3 mutually perpendicular planes as required.

4.8.3 Government provisions.- Test equipment listed in 4.8.1.1.1 will be used for Government product inspection (reference 4.3(b)), and to settle disputes (reference 4.2.1).

4.9 Test methods and procedures.

4.9.1 Environmental.

4.9.1.1 Vibration.- This test shall be conducted in accordance with method 201 of Standard MIL-STD-202. Performance measurements of the generator shall be made after vibration. Subsequent to vibration, the generator shall be subjected to a visual-tactile examination for damage. The generator shall then be subjected to the tests specified in 4.9.1.2 to 4.9.3.6 inclusive.

4.9.1.2 Operating temperature.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the generator is thermally stabilized at each indicated temperature, it shall be subjected to the tests specified in 4.9.3 and shall meet the requirements specified.

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4.9.1.3 Storage temperature.- This test shall be conducted in accordance with Procedure II, condition "A" and "H" of Specification MIL-F-13926. Upon completion of the specified exposure, the generator shall be inspected for damage and then subjected to the tests specified in 4.9.3, and shall meet the requirements specified. This test shall be performed on the production model only.

4.9.2 Electrical.

4.9.2.1 Circuitry.- A continuity check of the generator circuits shall be made using standard measuring equipment to determine conformance to the requirements specified.

4.9.2.2 Dielectric strength.- The generator shall be tested for dielectric strength in accordance with method 301 of Standard MIL-STD-202, and shall meet the requirements specified.

4.9.2.3 Insulation resistance.- This test shall be conducted in accordance with method 302, test condition "B", of Standard MIL-STD-202. The generator shall meet the requirements specified. This test shall immediately follow the dielectric strength test.

4.9.3 Performance.- The following test shall be conducted with the test equipment specified in 4.8.1.1.1 connected as shown on figure 4, or as described herein.

4.9.3.1 Balance, 4-kilocycle (kc).

4.9.3.1.1 Potentiometer R20 shall be adjusted so that a null occurs at TP1, using item 3, when the resolver is set at zero.

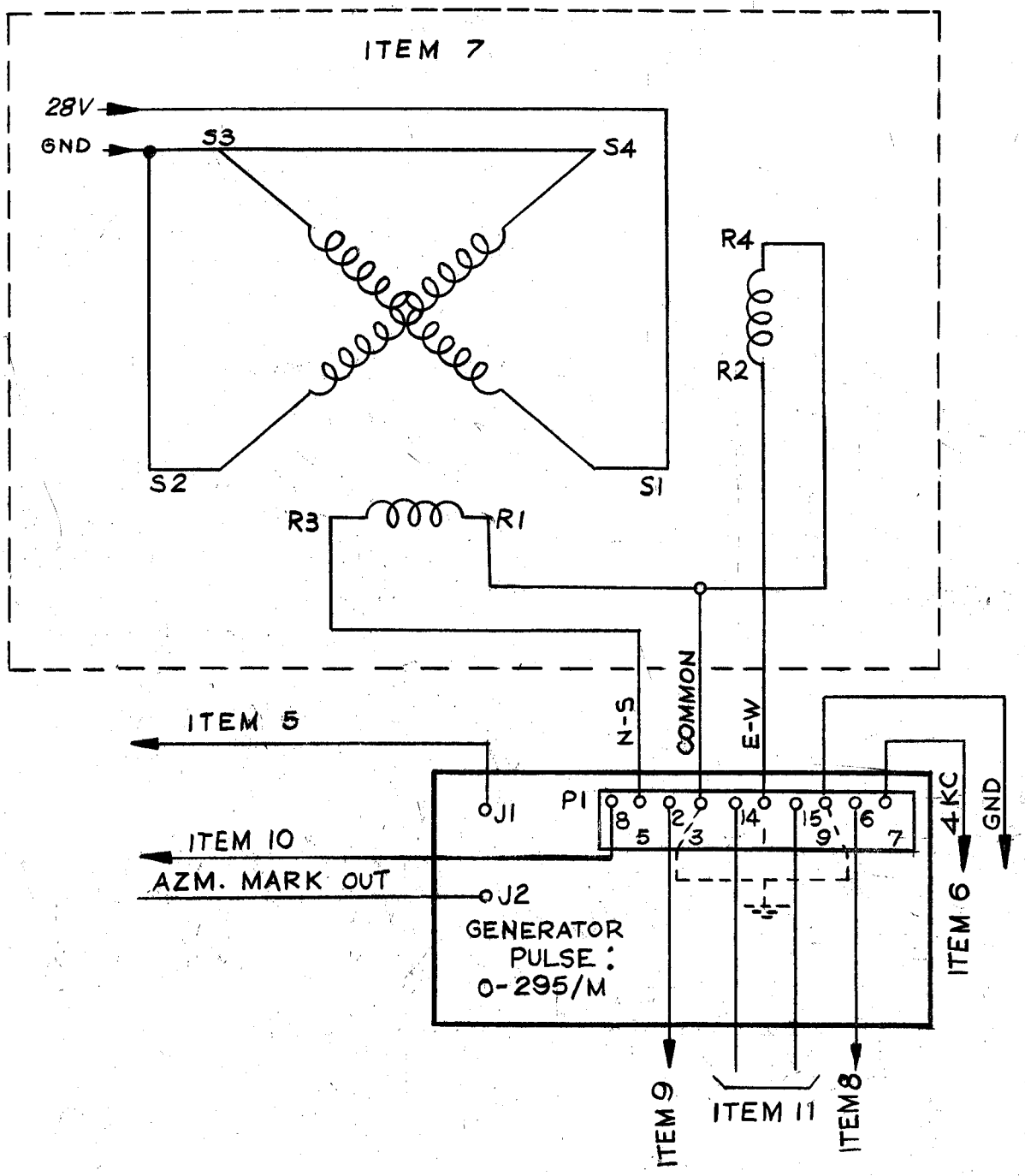
4.9.3.1.2 The reference carrier component combined with north-south (N-S) (terminal 5 of P1) resolver components shall give a resultant signal which varies as specified.

4.9.3.1.3 The residual voltage at the null shall not exceed the value specified.

4.9.3.2 Azimuth gate.

4.9.3.2.1 Voltmeter item 1 shall be connected from the cathode (pin 8) of electron tube V3B to ground.

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TEST SET-UP

FIGURE 4

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4.9.3.2.2 The output voltage shall vary as indicated in figure 2 when the resolver is rotated through a full revolution. An observation shall be made to determine that the gate has a width at least 30 degrees at the minus 36 volt points, 16 degrees plus or minus 3 degrees at the minus 16 volt points, and a minimum positive peak voltage of 1.0 volts all centered about the resolver null.

4.9.3.2.3 The peak voltage occurring at approximately 180 degrees shall be negative with a minimum amplitude of 50 volts.

4.9.3.2.4 The 4 kilocycle component, as measured with voltmeter item 4, at the cathode of V3B, shall not exceed 1.0 volt at any point within the 30 degrees area tested above.

4.9.3.3 Azimuth mark.

4.9.3.3.1 Electron tube voltmeter item 2 shall be connected from the suppressor grid (pin 7) of V7 to ground.

4.9.3.3.2 Potentiometer R10 shall be set in its full clockwise position, and the output voltage shall vary as indicated in figure 3 when the resolver is turned one full revolution. It shall be determined that the gate has a width of 3.5 degrees plus or minus 1 degree at the minus 8 volt points and a peak voltage more positive than minus 1 volt.

4.9.3.3.3 The peak voltage occurring at approximately 180 degrees shall be negative with the minimum amplitude of 20 volts.

4.9.3.3.4 The potentiometer R10 shall be set in its full counter-clockwise position and the output voltage shall vary as indicated in figure 3. It shall be determined that the gate has a width 0.8 degrees plus or minus 0.1 degree at the minus 8 volt points and a peak voltage of more than minus 3 volts.

4.9.3.4 Azimuth mark triggering level.

4.9.3.4.1 The oscilloscope item 3 shall be connected from J2 to ground.

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4.9.3.4.2 The voltmeter item 2 shall be connected as in 4.9.3.3.1 and starting at minus 3 degrees the resolver shall be rotated toward 0 degrees until the firing point of the multivibrator V6 is as indicated by the presence of the azimuth pulse at J2.

4.9.3.4.3 The voltmeter item 2 reading, when the multivibrator reaches the firing point, shall be negative and not more than 10 volts in amplitude.

4.9.3.5 Azimuth mark repetition rate, length and amplitude.

4.9.3.5.1 The oscilloscope item 3 shall be connected as in 4.9.3.4.1 and pin 7 of V7 grounded.

4.9.3.5.2 The potentiometer R32 shall be varied between its limits and this action should vary the azimuth mark length over a minimum range of 500 to 750 microseconds.

4.9.3.5.3 Adjustment of the pulse length to 610 microseconds shall be accomplished by applying the trigger pulses which occur every 667 microseconds, and the oscilloscope markers (10 microseconds) simultaneously noting the time difference between the end of the azimuth mark pulse and pre-knock pulse. The azimuth mark amplitude shall be a minimum of 6 volts.

4.9.3.6 Output at 80 revolutions per minute.

4.9.3.6.1 The resolver shall be rotated at 80 plus or minus 5 revolutions per minute with potentiometer R10 in its maximum clockwise position. The azimuth mark shall appear at J2 once each revolution of the resolver.

4.9.4 Identification markings.- Markings shall be visually inspected for conformance to Standard MIL-STD-130.

4.9.5 Final acceptance.- Final acceptance of a lot of generators shall be withheld by the Government inspector until the sample units, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

4.10 Preservation, packaging, packing and marking.- Quality assurance provisions for preservation, packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The generator shall be cleaned, preserved, packed, and marked in accordance with Specification MIL-F-14619 P3916. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The generator is intended to receive a 4-kc reference carrier of constant amplitude and phase and two 4-kc voltages, which vary in amplitude and phase as the sine and cosine of the associated resolver positions. Using the 4-kc inputs and a positive 25-volt, 1 u sec, preknock pulse (at 1000 or 1500 pulses), the generator develops one azimuth mark per revolution of the associated resolvers and a azimuth gate centered about the mark. The azimuth mark and gate are used in the acquisition indicator display. The generator is part of the precision indicator located in either the tactical control console or the tracking console and is used with the M33 Fire Control System, and Radar Course Directing Central AN/MSW-1 and AN/MSW-3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Definitions.

6.3.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.3.2 Tactile.- Skillful manipulation or sensing.

6.4 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

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